Do Transformers Learn Internal World Models?

A small grid-world experiment with a tiny transformer

The question

I trained a small transformer to predict the next move of an agent walking on a 5×5 grid. I never told it where the agent was. I wanted to know whether the model works out the agent's position on its own.

What I did

- Made 30,000 random walks of 20 moves each. The agent only makes legal moves, so it never leaves the grid.
- Trained a 3-layer transformer to predict the next move. It sees only the moves, never the positions.
- Trained simple linear probes on the model's hidden states, one per layer, to guess the agent's cell. If a basic probe can read the position, the model is storing it.
- Added two controls: an untrained model of the same size, and probes trained on shuffled position labels.

Probes were trained on 2,000 walks and tested on 3,000 walks the model had not seen.

What I found

The model learned the game. Validation loss was 1.186, close to the best possible (about 1.19) for random legal moves. Its top predicted move never left the grid.

Position can be read from the hidden states. This is the probe accuracy by layer. Always guessing the most common cell scores about 0.05.

Layer	Trained model	Untrained model	Shuffled labels
Embedding	0.096	0.096	0.048
Layer 1	0.473	0.181	0.050

Layer	Trained model	Untrained model	Shuffled labels
Layer 2	0.971	0.195	0.058
Layer 3	0.929	0.198	0.052

The trained model reaches 97% at layer 2. The untrained model stays near 20%, and shuffled labels stay at the baseline. So the result comes from training, and the probe is not just memorizing. (Trained and shuffled used a 1000-step probe, untrained used 500 steps.)

Different paths to the same cell look alike. I compared the hidden states of walks that ended in the same cell with walks that ended in different cells. The gap between the two grows with depth in the trained model and stays flat in the untrained one.

Layer	Trained	Untrained
Embedding	0.07	0.07
Layer 1	0.14	0.07
Layer 2	0.37	0.07
Layer 3	0.44	0.07

Walks ending in the same cell reach a similarity of 0.57, not 1.0, so the model also remembers some of the path.

What this means

The model seems to keep track of where the agent is, even though it was only trained to predict moves. But I only showed that the information is there, not that the model actually uses it.

Limits

- No causal test, so readable does not mean used.
- One run, one grid size, no obstacles. I did not fix the training seed, so exact numbers will change a little between runs.
- Layer 3 is a bit lower than layer 2 (0.929 vs 0.971). A longer probe narrowed the gap, but I did not find out whether the rest is real.

- Probes were trained on only 2,000 walks.

Next steps

Test whether the model uses the position by editing the position direction and watching its predictions. Repeat with several seeds, then try obstacles and bigger grids.